Tutorial - 4

(2) ~~Let~~ This is the Hamiltonian of a SHO with an extra linear term in $\hat{x}$. We do a coordinate shift to make this linear term = 0.

Let $\hat{x} = \hat{y} - x_0$. $\Rightarrow$ $\hat{p}_y = \hat{p}_x$

$$\therefore \frac{K}{2}(y - x_0)^2 + \frac{e\phi_0}{L}(y - x_0)$$

$$= \frac{K}{2}(y^2 + x_0^2 - 2x_0 y) + \frac{e\phi_0}{L}(y - x_0)$$

$$= \frac{Ky^2}{2} + \frac{Kx_0^2}{2} - Kx_0 y + \frac{e\phi_0}{L} y - \frac{e\phi_0}{L} x_0.$$

The ~~b~~ term linear in $\hat{y}$

$$-kx_0 + \frac{e\phi_0}{L} = 0.$$

$$\Rightarrow \boxed{x_0 = \frac{e\phi_0}{kL}} \quad \text{--- (1)}$$

Replacing in H:

$$H = \frac{\hat{p}^2}{2m} + \frac{ky^2}{2} + \frac{kx_0^2}{2} - \frac{e\phi_0}{L}x_0$$

$$= \frac{p^2}{2m} + \frac{ky^2}{2} + \frac{k}{2}\frac{e^2\phi_0^2}{k^2L^2} - \frac{e\phi_0}{L}\cdot\frac{e\phi_0}{kL^2}$$

$$\boxed{\hat{H} = \frac{\hat{p}^2}{2m} + \frac{k\hat{y}^2}{2} - \frac{e^2\phi_0^2}{2kL^2}}$$

This is a ~~SH~~ the Hamiltonian of a SHO with

$$\boxed{\omega = \sqrt{\frac{k}{m}}}$$

& $$\boxed{E_0 = \frac{\hbar\omega}{2} - \frac{e^2\phi_0^2}{2kL^2}}$$ (Ans)

Ground state wavef^n: ~~for~~ ~~no~~ $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}(x-x_0)^2}$

(Ans) $x_0 = \frac{e\phi_0}{kL}$

③ ~~$\psi(x)$~~. $\langle x\rangle = \frac{1}{2}\left\langle \langle 0| + \langle 1| \right| \hat{x} \left| |1\rangle + |0\rangle \right\rangle$.

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\,(a_+ + a_-).$$

$$\therefore \langle x\rangle = \frac{1}{2}\left\langle \langle 0| + \langle 1| \right| a_+ + a_- \left| |1\rangle + |0\rangle \right\rangle \sqrt{\frac{\hbar}{2m\omega}}\,\frac{1}{2} \quad (c)$$

$$= c\left[\left\langle \langle 0| + \langle 1| \right| a_+ \left| |0\rangle + |1\rangle \right\rangle + \left\langle \langle 0| + \langle 1| \right| a_- \left| |0\rangle + |1\rangle \right\rangle\right]$$

$$= c\left[\left\langle \langle 0| + \langle 1| \right| A_1 |1\rangle + A_2 |2\rangle \right\rangle + \left\langle \langle 0| + \langle 1| \right| A_0 |0\rangle \right\rangle\right]$$

$$= c\left[A_1\langle 1|1\rangle + A_0\langle 0|0\rangle\right] = 2c.$$

Now, $a_+|0\rangle = |1\rangle \Rightarrow A_1 = 1.$

$a_-|1\rangle = |0\rangle \Rightarrow A_0 = 1.$

$$\therefore \boxed{\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}}}$$

$$\therefore \hat{x}^2 = \frac{\hbar}{2m\omega}(a_+ + a_-)^2$$

$$= \frac{\hbar}{2m\omega}(a_+^2 + a_-^2 + a_+a_- + a_-a_+).$$

$$[a_+, a_-] = -1 \Rightarrow a_-a_+ = a_+a_- + 1.$$

$$\hat{x}^2 = \frac{\hbar}{2m\omega}(a_+^2 + a_-^2 + 2a_+a_- + 1).$$

$$\therefore \langle \hat{x}^2 \rangle = \langle \langle 0| + \langle 1| \; a_+^2 + a_-^2 + 2a_+a_- + 1 \; |0\rangle + |1\rangle \rangle \; \frac{\hbar}{2m\omega} \rightarrow c$$

$$= \Big[\langle \langle 0| + \langle 1| \; a_+^2 \; |0\rangle + |1\rangle \rangle$$

$$+ \langle \langle 0| + \langle 1| \; a_-^2 \; |0\rangle + |1\rangle \rangle$$

$$+ 2\langle \langle 0| + \langle 1| \; a_+a_- \; |0\rangle + |1\rangle \rangle$$

$$+ \langle \langle 0| + \langle 1| \; 1 \; |0\rangle + |1\rangle \rangle \Big] c$$

$$= \Big[\langle \langle 0| + \langle 1| \; A_2|2\rangle + A_3|3\rangle \rangle$$

$$+ \langle \langle 0| + \langle 1| \; 0\rangle + |1\rangle \rangle$$

$$+ 2\langle \langle 0| + \langle 1| \; A_1|1\rangle \rangle \Big] c$$

$$= \langle 0|0\rangle + \langle 1|1\rangle + 2$$

$$= 2$$

$$= (2A_1 + 2)c$$

Now, $a_+a_-|1\rangle = a_+|0\rangle = |1\rangle \Rightarrow A_1 = 1$

$$\therefore \boxed{\langle \hat{x}^2 \rangle = 4c = \frac{3\hbar}{2m\omega} = \frac{\hbar}{m\omega}}$$

$$\therefore \Delta x = \sqrt{\frac{\hbar}{m\omega} - \frac{\hbar}{2m\omega}} = \sqrt{\frac{\hbar}{2m\omega}}.$$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}_+ - \hat{a}_-).$$

$$\therefore \langle p \rangle = \langle \langle 0| + \langle 1| \; \hat{a}_+ - \hat{a}_- \; |1\rangle + |0\rangle \rangle \; \frac{i}{2}\sqrt{\frac{\hbar m\omega}{2}} \rightarrow c.$$

$$= c\Big[\langle \langle 0| + \langle 1| \; a_+ \; |1\rangle + |0\rangle \rangle$$

$$- \langle \langle 0| + \langle 1| \; a_- \; |1\rangle + |0\rangle \rangle \Big]$$

$$= c\Big[\langle \langle 0| + \langle 1| \; A_2|2\rangle + A_1|1\rangle \rangle - \langle \langle 0| + \langle 1| \; A_0|0\rangle \rangle \Big] c$$

$\Rightarrow \langle p \rangle = c[A_1 \langle 1|1\rangle - A_0 \langle 0|0\rangle]$

$\boxed{\langle p \rangle = 0}$ $\quad [A_1 = A_0 = 1]$.

$\therefore$ 6. $\hat{p}^2 = -\frac{m\omega\hbar}{2}(a_+^2 + a_-^2 - a_+a_- - a_-a_+)$

$= -\frac{m\omega\hbar}{2}(a_+^2 + a_-^2 - 2a_+a_- - 1)$

$\therefore \langle \hat{p}^2 \rangle = \langle \langle 0| + \langle 1| \, | a_+^2 + a_-^2 - 2a_+a_- - 1 | \, |0\rangle + |1\rangle \rangle$

$-\frac{m\omega\hbar}{4} \rightarrow c$

$= c[\langle \langle 0| + \langle 1| \, a_+^2 \, |0\rangle + |1\rangle \rangle +$

$a_- \langle \langle 0| + \langle 1| \, a_-^2 \, |1\rangle + |0\rangle \rangle$

$- 2 \langle \langle 0| + \langle 1| \, a_+a_- \, |0\rangle + |1\rangle \rangle -$

$\langle \langle 0| + \langle 1| \, |1\rangle + |0\rangle \rangle]$

$= c[\cancel{\langle \langle 0| + \langle 1| \, A_2 |2\rangle + A_3 |3\rangle}$

$-2A_1 - 2]$. (from previous derivation for $\langle x^2 \rangle$)

$= -4c \quad [A_1 = 1]$

$\boxed{\langle p^2 \rangle = \hbar m\omega}$

$\therefore \Delta p = \sqrt{\hbar m\omega}$

$\therefore \Delta x \, \Delta p = \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\hbar m\omega} = \frac{\hbar}{\sqrt{2}} > \frac{\hbar}{2}$

Hence, uncertainty principle is verified.

① $\hat{x} = i\hbar \frac{\partial}{\partial p}$ — (momentum-basis)

$\therefore [\hat{x}, F(\hat{p})]\psi = [i\hbar \frac{\partial}{\partial p}(F(\hat{p})\psi) - F(\hat{p}) \, i\hbar \frac{\partial}{\partial p}\psi]$.

$= i\hbar [\frac{\partial F(\hat{p})}{\partial p}\psi + F(\hat{p})\frac{\partial \psi}{\partial p} - F(\hat{p})\frac{\partial \psi}{\partial p}]$.

$= i\hbar \frac{\partial F}{\partial p}\psi$

$\Rightarrow \boxed{[\hat{x}, F(\hat{p})] = i\hbar \frac{\partial F}{\partial p}}$ — (1).

Similarly $\hat{p} = -i\hbar \frac{\partial}{\partial x}$, & $[\hat{p}, G(\hat{x})] = -i\hbar \frac{\partial G}{\partial x}$ — (2)

$$\frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[A, H] + \frac{\partial A}{\partial t}$$ — Heisenberg's EOM.

Putting $\hat{A} = \hat{x}$; $\frac{\partial \hat{x}}{\partial t} = 0$

$$[\hat{x}, \hat{H}] = \left[x, \frac{\hat{p}^2}{2m} + V(x)\right].$$

$$= \frac{1}{2m}[x, \hat{p}^2] + [x, V(x)]$$

$$= \frac{1}{2m}(2i\hbar\hat{p}) = \frac{i\hbar\hat{p}}{m}.$$

$$\therefore \boxed{\frac{d\hat{x}_H}{dt} = \frac{\hat{p}}{m}} \quad \text{— (1)}$$

Putting $\hat{A} = \hat{p}$; $\frac{\partial \hat{p}}{\partial t} = 0.$

$$\therefore [\hat{p}, \hat{H}] = \left[p, \frac{\hat{p}^2}{2m} + V(x)\right] = [\hat{p}, V(x)]$$

$$\therefore [\hat{p}, V(x)]\psi = -i\hbar\left[\frac{\partial}{\partial x}(V\psi) - V\frac{\partial \psi}{\partial x}\right].$$

$$= -i\hbar\left[V\frac{\partial \psi}{\partial x} + \psi\frac{\partial V}{\partial x} - V\frac{\partial \psi}{\partial x}\right].$$

$$= -i\hbar\frac{\partial V}{\partial x}\psi.$$

$$\Rightarrow [\hat{p}, V(x)] = -i\hbar\frac{\partial V}{\partial x}.$$

$$\therefore \boxed{\frac{d\hat{p}_H}{dt} = -\frac{\partial V}{\partial \hat{x}}(\hat{x}_H)} \quad \text{— (II)}$$

From (1) we get

$$\hat{p}_H = m\frac{d\hat{x}_H}{dt}.$$

$$\Rightarrow \frac{d\hat{p}_H}{dt} = \boxed{m\frac{d^2\hat{x}_H}{dt^2} = -\frac{\partial V(\hat{x}_H)}{\partial \hat{x}}} \quad \text{— (3)}$$